

The Signed Schur Positivity of ∇m_μ

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Main result

Theorem (Qiu–Zhang, 2026)

For any partitions $\mu, \lambda \vdash n$ and any integer $r \geq 1$,

$$(-1)^{|\mu| - \ell(\mu)} \langle \nabla^r m_\mu, s_\lambda \rangle \in \mathbb{N}[q, t].$$

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In particular, the case $r = 1$ proves Conjecture IV of Bergeron–Garsia–Haiman–Tesler.

- The sign depends only on the monomial input m_μ .
- The conclusion holds simultaneously for every Schur coefficient.
- The proof also gives a shifted e-positive companion theorem.

D. Qiu and M. Zhang, *The Schur positivity of ∇m_μ* , arXiv:2607.00940.

From P_μ to the integral form J_μ

Macdonald's monic orthogonal basis is naturally defined over $\mathbb{Q}(q, t)$:

$$P_\mu = m_\mu + \sum_{\lambda < \mu} u_{\lambda\mu}(q, t) m_\lambda, \quad u_{\lambda\mu}(q, t) \in \mathbb{Q}(q, t).$$

In general these coefficients have denominators in q and t . Define

$$c_\mu(q, t) = \prod_{c \in \mu} (1 - q^{a(c)} t^{\ell(c)+1}), \quad J_\mu = c_\mu P_\mu \in \Lambda_{\mathbb{Z}[q, t]}.$$

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What this normalization does—and does not do

It clears the rational denominators, but does not give Schur positivity: $J_{(1)} = (1 - t)s_{(1)}$.

Macdonald, integral form of the Macdonald polynomials.

The modification of J_μ reveals Schur positivity

The modified Macdonald polynomial is obtained by a plethystic change:

$$\boxed{\tilde{H}_\mu[X; q, t] = t^{n(\mu)} J_\mu \left[\frac{X}{1 - t^{-1}}; q, t^{-1} \right]}, \quad n(\mu) = \sum_{i \geq 1} (i - 1) \mu_i.$$

Its Schur expansion is

$$\tilde{H}_\mu[X; q, t] = \sum_{\lambda \vdash |\mu|} \tilde{K}_{\lambda\mu}(q, t) s_\lambda.$$

Macdonald positivity

Haiman proved $\tilde{K}_{\lambda\mu}(q, t) \in \mathbb{N}[q, t]$.

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Thus the three normalizations have different purposes:

$$\underbrace{P_\mu}_{\text{monic and orthogonal}} \longrightarrow \underbrace{J_\mu}_{\text{integral}} \longrightarrow \underbrace{\tilde{H}_\mu}_{\text{Schur-positive}}.$$

The eigenvalue T_μ is read from the diagram

The modified Macdonald polynomials form the eigenbasis of ∇ :

$$\nabla \tilde{H}_\mu[X; q, t] = T_\mu \tilde{H}_\mu[X; q, t], \quad T_\mu = \prod_{c \in \mu} q^{a'(c)} t^{\ell'(c)} = q^{n(\mu')} t^{n(\mu)}.$$

For a cell in column j and row i , counted from the south-west corner,
 $a'(c) = j - 1$ and $\ell'(c) = i - 1$.

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For a cell in column j and row i , counted from the south-west corner, $a'(c) = j - 1$ and $\ell'(c) = i - 1$.

t	qt	
1	q	q^2

$$\mu = (3, 2), \quad T_{(3,2)} = (1)(q)(q^2)(t)(qt) = q^4 t^2.$$

In degree 3, the eigenvalues needed below are

$$T_{(3)} = q^3, \quad T_{(2,1)} = qt, \quad T_{(1,1,1)} = t^3.$$

Example: changing basis for $m_{(2,1)}$

In degree 3,

$$\tilde{H}_{(3)} = s_{(3)} + (q + q^2)s_{(2,1)} + q^3s_{(1,1,1)},$$

$$\tilde{H}_{(2,1)} = s_{(3)} + (q + t)s_{(2,1)} + qt s_{(1,1,1)},$$

$$\tilde{H}_{(1,1,1)} = s_{(3)} + (t + t^2)s_{(2,1)} + t^3s_{(1,1,1)}.$$

Since $m_{(2,1)} = s_{(2,1)} - 2s_{(1,1,1)}$, inversion gives, with

$$R = q^2 + qt + t^2 + 2q + 2t + 2,$$

$$\begin{aligned} m_{(2,1)} = & -\frac{t+2}{(q-t)(q^2-t)} \tilde{H}_{(3)} + \frac{R}{(q^2-t)(q-t^2)} \tilde{H}_{(2,1)} \\ & - \frac{q+2}{(q-t)(q-t^2)} \tilde{H}_{(1,1,1)}. \end{aligned}$$

Example: applying ∇ to $m_{(2,1)}$

Multiply the three modified Macdonald components by q^3 , qt , and t^3 , respectively:

$$(\tilde{H}_{(3)}, \tilde{H}_{(2,1)}, \tilde{H}_{(1,1,1)}) \xrightarrow{\nabla} (q^3 \tilde{H}_{(3)}, qt \tilde{H}_{(2,1)}, t^3 \tilde{H}_{(1,1,1)}).$$

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Returning to the Schur basis, all rational denominators cancel:

$$\nabla m_{(2,1)} = -2s_{(3)} - A(q, t)s_{(2,1)} - B(q, t)s_{(1,1,1)},$$

where

$$A(q, t) = qt(q + t + 2) + 2(q^2 + q + t^2 + t),$$

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$$B(q, t) = qt(q^2 + qt + t^2) + 2(q^3 + q^2t + qt^2 + qt + t^3).$$

Since $|\mu| - \ell(\mu) = 3 - 2 = 1$,

$$\boxed{-\nabla m_{(2,1)} = 2s_{(3)} + A(q, t)s_{(2,1)} + B(q, t)s_{(1,1,1)}}, \quad A, B \in \mathbb{N}[q, t].$$

What Schur positivity and e -positivity mean

With the Hall inner product,

$$\langle s_\lambda, s_\mu \rangle = \delta_{\lambda, \mu}.$$

Thus

$$[s_\lambda]f = \langle f, s_\lambda \rangle, \quad [e_\lambda]f = \langle f, \omega(m_\lambda) \rangle.$$

A symmetric function is Schur-positive, respectively e -positive, when all coefficients in the corresponding basis lie in $\mathbb{N}[q, t]$.

- Schur positivity resembles a decomposition into irreducibles.
- e -positivity is stronger and needs a different coefficient extractor.

Diagonal harmonics is the model case

Theorem (Haiman)

The bigraded Frobenius characteristic of diagonal harmonics is

$$\text{Frob}_{q,t}(DH_n) = \nabla e_n.$$

Consequently,

$$\langle \nabla e_n, s_\lambda \rangle \in \mathbb{N}[q, t].$$

Since $e_n = m_{(1^n)}$, this is the monomial case $\mu = (1^n)$.

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Representation-theoretic meaning

The coefficient of s_λ is the bigraded multiplicity of the irreducible representation indexed by λ .

The BGHT conjecture and the natural sign

Following the paper, write

$$F_\mu := (-1)^{|\mu| - \ell(\mu)} m_\mu.$$

BGHT, Conjecture IV (1999)

For all partitions $\mu, \lambda \vdash n$,

$$\langle \nabla F_\mu, s_\lambda \rangle \in \mathbb{N}[q, t].$$

For a single part,

$$F_n := F_{(n)} = (-1)^{n-1} p_n.$$

The notation F_μ will also be used in the labelled-multiset recursion.

Bergeron–Garsia–Haiman–Tesler, Methods Appl. Anal. 6 (1999), Conjecture IV.

Previous work

- $\mu = (1^n)$: diagonal harmonics and Haiman's theorem.
- $\mu = (n)$: the square-path theorem, conjectured by Loehr–Warrington and proved by Sergel.
- Hook partitions: results and parking-function models of Sergel.
- $\mu = (2^k, 1^\ell)$: a positive C -operator recursion of Qu–Xin.

The last result suggested the general strategy:

signed monomial $\longrightarrow$ positive expansion in $C_\alpha(1)$.

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Arbitrary partitions require controlling unequal and repeated parts.

Haiman; Loehr–Warrington; Sergel; Qu–Xin.

The C -operators

Haglund–Morse–Zabrocki define

$$C_a(f)[X] = \left(-\frac{1}{q}\right)^{a-1} \langle z^a \rangle f \left[X - \frac{1 - 1/q}{z} \right] \sum_{r \geq 0} h_r[X] z^r.$$

For a composition $\alpha = (\alpha_1, \dots, \alpha_m)$,

$$C_\alpha(f) := C_{\alpha_1}(C_{\alpha_2}(\cdots C_{\alpha_m}(f) \cdots)).$$

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$$C_\alpha(f) := C_{\alpha_1}(C_{\alpha_2}(\dots C_{\alpha_m}(f) \dots)).$$

A fundamental identity is

$$e_n = \sum_{\alpha \models n} C_\alpha(1).$$

Haglund–Morse–Zabrocki, *Canad. J. Math.* 64 (2012), 822–844.

Shuffle theory gives output positivity

For every composition $\alpha \models n$, partition $\lambda \vdash n$, and integer $r \geq 1$,

$$\langle \nabla^r C_\alpha(1), s_\lambda \rangle \in \mathbb{N}[q, t].$$

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The mechanism is

$\nabla^r C_\alpha(1) \longrightarrow$ nonnegative sum of column LLT polynomials $\longrightarrow$ Schur-positive.

- $r = 1$: the compositional shuffle theorem.
- $r \geq 1$: Mellit's compositional (km, kn) -shuffle theorem.

Carlsson–Mellit; Mellit; Grojnowski–Haiman.

Two certificates for the same coefficient

Using $F_\mu = (-1)^{|\mu|-\ell(\mu)} m_\mu$,

$$\langle \nabla^r F_\mu, s_\lambda \rangle = (-1)^{|\mu|-\ell(\mu)} \langle \nabla^r m_\mu, s_\lambda \rangle.$$

The proof establishes independently that

$$\langle \nabla^r F_\mu, s_\lambda \rangle \in \mathbb{Z}[q, t], \quad \langle \nabla^r F_\mu, s_\lambda \rangle \in \mathbb{Q}_{\geq 0}[q, t].$$

Therefore

$$\langle \nabla^r F_\mu, s_\lambda \rangle \in \mathbb{Z}[q, t] \cap \mathbb{Q}_{\geq 0}[q, t] = \mathbb{N}[q, t].$$

The two statements come from different expansions and control different features.

BHMPS and inverse Kostka give integrality

- ❶ Blasiak–Haiman–Morse–Pun–Seelinger give an LLT expansion of $\nabla^r s_\nu$ with coefficients in $\mathbb{Z}[q, t]$. Hence

$$\langle \nabla^r s_\nu, s_\lambda \rangle \in \mathbb{Z}[q, t].$$

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- ❷ The inverse Kostka transition is integral:

$$m_\mu = \sum_{\nu \vdash n} c_{\mu\nu} s_\nu, \quad c_{\mu\nu} \in \mathbb{Z}.$$

Therefore

$$\langle \nabla^r m_\mu, s_\lambda \rangle = \sum_{\nu \vdash n} c_{\mu\nu} \langle \nabla^r s_\nu, s_\lambda \rangle \in \mathbb{Z}[q, t].$$

Consequently,

$$\boxed{\langle \nabla^r F_\mu, s_\lambda \rangle \in \mathbb{Z}[q, t]}.$$

This proves integrality, but not coefficientwise nonnegativity.

Blasiak–Haiman–Morse–Pun–Seelinger, JAMS 38 (2025); Eġecioġlu–Remmel, Linear Multilinear Algebra 26 (1990).

The augmented signed monomial $\tilde{F}_A$

Let A be a labelled multiset of positive integers. Set

$$S_A := \sum_{x \in A} x, \quad \ell(A) := |A|, \quad \mu(A) := \text{the partition of its values.}$$

Following the paper, define

$$F_A := (-1)^{S_A - \ell(A)} m_{\mu(A)}, \quad \tilde{F}_A := \left(\prod_{i \geq 1} m_i(A)! \right) F_A.$$

Here $m_i(A)$ is the multiplicity of the value i in A . Thus, if $\mu(A) = \mu$, then

$$\tilde{F}_A = \left(\prod_{i \geq 1} m_i(\mu)! \right) F_\mu.$$

The correction term $\Theta_{A,c}$

Let $c \geq 1$ and suppose that

$$x \geq c \quad \text{for every } x \in A.$$

Define

$$\Theta_{A,c} := \tilde{F}_{A \cup \{c\}} - \sum_{j=1}^{c-1} q^{j-1} C_j(\tilde{F}_{A \cup \{c-j\}}).$$

Equivalently,

$$\tilde{F}_{A \cup \{c\}} = \Theta_{A,c} + \sum_{j=1}^{c-1} q^{j-1} C_j(\tilde{F}_{A \cup \{c-j\}}).$$

The continuation terms keep the same number of labelled elements, but replace the minimum c by $c - j < c$.

The main recursion

If $x \geq c$ for every $x \in A$, then

$$\Theta_{A,c} = \sum_{T \subseteq A} \left(\sum_{b=1}^{S_T} \alpha_{T,b} C_{c+S_T-b}(\tilde{F}_{(A \setminus T) \cup \{b\}}) + \beta_T C_{c+S_T}(\tilde{F}_{A \setminus T}) \right).$$

Recall that

$$\Theta_{A,c} := \tilde{F}_{A \cup \{c\}} - \sum_{j=1}^{c-1} q^{j-1} C_j(\tilde{F}_{A \cup \{c-j\}}).$$

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Recall that

$$\Theta_{A,c} := \tilde{F}_{A \cup \{c\}} - \sum_{j=1}^{c-1} q^{j-1} C_j(\tilde{F}_{A \cup \{c-j\}}).$$

Every nonzero inner $\tilde{F}$ has fewer labelled elements than $\tilde{F}_{A \cup \{c\}}$, and

$$\alpha_{T,b}, \beta_T \in \mathbb{N}[q].$$

Qiu-Zhang, main recursion theorem.

The coefficients lie in $\mathbb{N}[q]$

For $T \neq \emptyset$,

$$\alpha_{T,b} = (|T| - 1)! \sum_{\substack{x \in T \\ b \leq x}} q^{(c+x-b-1) \bmod x}, \quad \alpha_{\emptyset,b} = 0.$$

Moreover,

$$\beta_T = (|T| + 1)! [c]_q + |T|! \sum_{x \in T} (q^c + q^{c+1} + \cdots + q^{x-1}).$$

Since $x \geq c$ for $x \in A$,

$$\alpha_{T,b}, \beta_T \in \mathbb{N}[q].$$

The positive C -cone

Define

$$\mathcal{C}^+ := \sum_{\alpha} \mathbb{N}[q] C_{\alpha}(1).$$

It is stable under every C_j :

$$C_j(C_{\alpha}(1)) = C_{(j,\alpha)}(1).$$

The positive $\mathcal{C}$ -cone

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It is stable under every C_j :

$$C_j(C_{\alpha}(1)) = C_{(j,\alpha)}(1).$$

The main recursion was designed to prove

$$\tilde{F}_A \in \mathcal{C}^+.$$

Its inner augmented functions are smaller, its coefficients lie in $\mathbb{N}[q]$, and every outer operator preserves the cone.

Positive C-expansion theorem

Theorem

For every labelled multiset A of positive integers,

$$\tilde{F}_A \in \sum_{\alpha \models S_A} \mathbb{N}[q] C_\alpha(1).$$

If $\mu(A) = \mu$, then

$$\tilde{F}_A = \left(\prod_{i \geq 1} m_i(\mu)! \right) F_\mu.$$

Therefore

$$F_\mu = \sum_{\alpha \models |\mu|} a_{\mu, \alpha} C_\alpha(1), \quad a_{\mu, \alpha} \in \mathbb{Q}_{\geq 0}[q].$$

The positivity certificate

Choose A with $\mu(A) = \mu$. Then

$$F_\mu = \sum_{\alpha \models n} a_{\mu, \alpha} C_\alpha(1), \quad a_{\mu, \alpha} \in \mathbb{Q}_{\geq 0}[q].$$

Applying ∇^r and extracting a Schur coefficient gives

$$\begin{aligned} \langle \nabla^r F_\mu, s_\lambda \rangle &= \sum_{\alpha \models n} a_{\mu, \alpha} \langle \nabla^r C_\alpha(1), s_\lambda \rangle \\ &\in \mathbb{Q}_{\geq 0}[q, t]. \end{aligned}$$

Positive C -expansion + compositional shuffle theorem + LLT Schur positivity.

Proof of the main theorem

For

$$\langle \nabla^r F_\mu, s_\lambda \rangle = (-1)^{|\mu| - \ell(\mu)} \langle \nabla^r m_\mu, s_\lambda \rangle,$$

we have proved

$$\langle \nabla^r F_\mu, s_\lambda \rangle \in \mathbb{Q}_{\geq 0}[q, t]$$

from the positive C -expansion and shuffle theory.

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$$\langle \nabla^r F_\mu, s_\lambda \rangle \in \mathbb{Q}_{\geq 0}[q, t]$$

from the positive C -expansion and shuffle theory.

On the other hand, BHMPs and inverse Kostka give

$$\langle \nabla^r F_\mu, s_\lambda \rangle \in \mathbb{Z}[q, t].$$

Hence

$$\langle \nabla^r F_\mu, s_\lambda \rangle \in \mathbb{N}[q, t].$$

Example: the first recursion step for $\mu = (2, 1)$

Since the parts are distinct,

$$\tilde{F}_{\{2,1\}} = F_{(2,1)} = -m_{(2,1)}.$$

Take $A = \{2\}$ and $c = 1$. Since the continuation sum is empty, the main recursion gives

$$-m_{(2,1)} = 2C_1(-m_{(2)}) + qC_2(m_{(1)}) + (2 + q)C_3(1).$$

Every inner input now has at most one part.

Example: the positive C -expansion

The smaller inputs have the expansions

$$-m_{(2)} = C_{(1,1)}(1) + (1+q)C_{(2)}(1), \quad m_{(1)} = C_{(1)}(1).$$

Substituting them into the first recursion step gives

$$\begin{aligned} -m_{(2,1)} = & 2C_{(1,1,1)}(1) + (2+2q)C_{(1,2)}(1) \\ & + qC_{(2,1)}(1) + (2+q)C_{(3)}(1). \end{aligned}$$

Thus $-m_{(2,1)} \in \mathcal{C}^+$ before ∇ is applied.

Example: Schur positivity

Applying ∇ and the compositional shuffle theorem gives

$$\nabla(-m_{(2,1)}) = 2s_{(3)} + A(q, t)s_{(2,1)} + B(q, t)s_{(1,1,1)},$$

where

$$\begin{aligned} A(q, t) &= qt(q + t + 2) + 2(q^2 + q + t^2 + t), \\ B(q, t) &= qt(q^2 + qt + t^2) \\ &\quad + 2(q^3 + q^2t + qt^2 + qt + t^3). \end{aligned}$$

Hence

$$2, A(q, t), B(q, t) \in \mathbb{N}[q, t].$$

Theorem

For any $\mu, \lambda \vdash n$ and $r \geq 1$,

$$(-1)^{|\mu|-\ell(\mu)} \langle (\nabla^r m_\mu)[X; q+1], \omega(m_\lambda) \rangle \in \mathbb{N}[q, t].$$

Here $f[X; q+1]$ means that the parameter q in f is replaced by $q+1$. Since

$$[e_\lambda]f = \langle f, \omega(m_\lambda) \rangle,$$

this is an e-positivity statement after the shift.

Qiu-Zhang; D'Adderio, J. Combin. Theory Ser. A 172 (2020), 105212.

Why column LLT polynomials are enough

Mellit's shuffle theorem gives

$$\nabla^r C_\alpha(1) = \sum_D u_{\alpha,D}(q, t) \text{LLT}_D, \quad u_{\alpha,D}(q, t) \in \mathbb{N}[q, t],$$

where the relevant LLT_D are column LLT polynomials.

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where the relevant LLT_D are column LLT polynomials.

Two target theorems can then be used:

$$\begin{array}{ll} \text{LLT}_D & \text{is Schur-positive,} \\ \text{LLT}_D[X; q+1] & \text{is e-positive.} \end{array}$$

Thus the source recursion is unchanged in the two positivity results.

Grojnowski–Haiman; D'Adderio; Alexandersson–Sulzgruber.

Extension to the Schiffmann algebra

Let $M = (1 - q)(1 - t)$, and let r, k be coprime positive integers. The action satisfies

$$f[-MX^{r,1}] \cdot 1 = \nabla^r f.$$

If $\mu \vdash n$ and $\lambda \vdash kn$, then

$$(-1)^{|\mu| - \ell(\mu)} \left\langle m_\mu[-MX^{r,k}] \cdot 1, s_\lambda \right\rangle \in \mathbb{N}[q, t].$$

The shifted e -positive analogue also holds. The source C -expansion is unchanged; Mellit's (km, kn) -shuffle theorem provides the new target map.

Burban–Schiffmann; Mellit; BHMPs; Qiu–Zhang.

Paper Conjecture 5.2

The theorem proves integral C -positivity only after augmentation:

$$\tilde{F}_\mu = \left(\prod_{i \geq 1} m_i(\mu)! \right) F_\mu \in \sum_{\alpha \models |\mu|} \mathbb{N}[q] C_\alpha(1).$$

The paper conjectures the stronger statement

$$F_\mu = (-1)^{|\mu| - \ell(\mu)} m_\mu \in \sum_{\alpha \models |\mu|} \mathbb{N}[q] C_\alpha(1).$$

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The paper conjectures the stronger statement

$$F_\mu = (-1)^{|\mu| - \ell(\mu)} m_\mu \in \sum_{\alpha \models |\mu|} \mathbb{N}[q] C_\alpha(1).$$

For $\mu = (2, 2)$, dividing the recursive expansion produces a coefficient $q/2$ on $C_{(1,2,1)}(1)$, but another expansion is

$$\begin{aligned} m_{(2,2)} = & C_{(1,1,1,1)}(1) + (1 + q + q^2 + q^3) C_{(1,1,2)}(1) \\ & + (1 + q) C_{(1,3)}(1) + (q^2 + 2q) C_{(2,2)}(1) + (1 + q) C_{(4)}(1). \end{aligned}$$

Proof outline I: two C -operator identities

For every $n \geq 1$,

$$F_n = \sum_{j=1}^{n-1} q^{j-1} C_j(F_{n-j}) + [n]_q C_n(1).$$

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The first identity expands a one-part function.

The second moves a remaining factor through an existing C -operator.

Both identities follow directly from the HMZ definition of C_a .

Proof outline II: subtract and regroup

Starting from

$$\tilde{F}_{A \cup \{c\}} = \sum_{T \subseteq A} |T|! \tilde{F}_{A \setminus T} F_{c+S_T},$$

the proof proceeds as follows:

- 1 expand F_{c+S_T} with the first C -identity;
- 2 move the remaining factors with the second identity;
- 3 subtract the continuation sum defining $\Theta_{A,c}$;
- 4 group terms by the final pair (T, b) ;
- 5 identify the totals as $\alpha_{T,b}$ and β_T .

After regrouping, all coefficients lie in $\mathbb{N}[q]$.

Proof outline III: lexicographic induction

Choose $c = \min(A)$ and write $A = A' \cup \{c\}$. Then

$$\tilde{F}_A = \Theta_{A',c} + \sum_{j=1}^{c-1} q^{j-1} C_j(\tilde{F}_{A' \cup \{c-j\}}).$$

Induct on

$$(\ell(A), \min(A))$$

in lexicographic order.

- Terms in $\Theta_{A',c}$ have fewer labelled parts.
- Continuation terms have the same length and smaller minimum.
- $\mathcal{C}^+$ is stable under every C_j .

Hence $\tilde{F}_A \in \mathcal{C}^+$.

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The same source expansion yields shifted e -positivity and the Schiffmann-algebra extension.

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